Indian Institute of Technology, Jodhpur

Chemistry (CYL1010)

Assignment-1

Department: Chemistry Date of submission 24.2.26; Time: 12:00 midnight

Q. 1. A monochromatic radiation is incident on a solution of 0.05 M of an absorbing substance. The intensity of the radiation is reduced to $1/4^{th}$ of the initial value after passing through 10 cm length of solution. Calculate ε of the substance.

Q. 2. A substance when dissolved in water at 10^{-3}M concentration absorbs 10% of an incident radiation in a path length of 1 cm. What should be concentration of the solution to absorb 90% of the incident radiation?

Q. 3. In a Beer- Lambert law cell, the aqueous solution of a substance of known concentration absorbs 10% of incident light. What fraction of incident light will be absorbed by the same solution in a cell five times as long?

Q. 4. Calculate no. of moles of HCl (g) produced by the absorption of one joule of radiant energy of wavelength 480 nm in the reaction H2(g) + Cl2 (g) $\rightarrow$ 2HCl (g) if the quantum yield of photochemical reaction is 1×10^{6}.

Q. 5. In an experiment to measure the quantum efficiency of a photochemical reaction, the absorbing substance was exposed to 490 nm light from a 100 W source for 45 minutes. The intensity of the transmitted light was 40% of the intensity of the incident light. As a result of irradiation , 0.344 mol of the absorbing substance decomposed. Find quantum yield.

Q.6. A. State the Beer-Lambert law. B. Using Beer-Lambert law deduce the relation between fluorescent intensity [$F(\lambda_{ex}, \lambda_{em})$] and intensity of the source radiation (I_0).

Q. 7. Reason out the following observation:

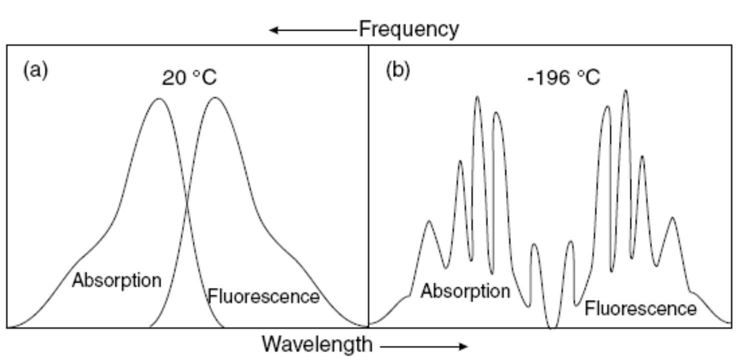

Figure (a) Absorption and fluorescence spectrum of a typical organic dye molecule in solution (a) at room temperature (20 °C) and (b) at −196 °C.

Q.8. State True or False with proper justification:

a) We observe spectral bands in UV-VIS spectrum rather than spectral lines.
b) Every molecule exhibits electronic transitions.
c) Fluorescence quantum yield can be greater than 1.
d) Phosphorescence is known as "delayed fluorescence"
e) Phosphorescence photons always emit from triplet ground state to singlet ground state
f) UV-VIS and Fluorescence spectroscopy measurements are done in continuous mode.

Q.9. Draw a neat diagram of an Fluorescence spectrophotometer and label all the components.

Q.10 Choose the correct option with proper justification:
[No justification will fetch no marks]

i) A molecule embedded in a rigid polymer matrix shows increased fluorescence. This is because:
A. ISC increases
B. Internal conversion decreases
C. Radiative decay stops
D. Absorption shifts

ii) If a molecule has:

- $k_r = 1 \times 10^8\ s^{-1}$, $k_{IC} = 4 \times 10^8\ \ s^{-1}$, $k_{ISC} = 5 \times 10^8\ \ s^{-1}$

Where k_r: denotes radiative emission rate and so on. The excited state depopulation is dominated by:
A. Fluorescence
B. Internal conversion
C. Intersystem crossing
D. Phosphorescence

iii) Two molecules A and B have same k_r, but molecule A has larger quantum yield (Φ) compared to B.
This indicates that A has:
A. A has larger absorption coefficient
B. A has smaller k_{nr}
C. A has larger Stokes shift
D. A has longer excitation wavelength

iv) A rigid molecule is converted into a strained non-planar structure through a specific synthetic route. It is observed that after modification fluorescence decreases and phosphorescence remains unchanged. Which rate constant is most likely increased?
A. k_f (rate of fluorescence)
B. k_{IC} (rate of internal conversion)

C. k_{ISC} (rate of intersystem crossing)
D. k_p (rate of phosphorescence)

v) A molecule absorbs 4×10^6 photons and emits only 2×10^5 photons as fluorescence photons due to ring strain–induced internal conversion.
What fraction of excited molecules decays non-radiatively?
A. 0.05
B. 0.50
C. 0.95
D. 0.80

Q.11. Draw Jablonski diagram and mark all the photophysical processes along with their timescale.
Q.12. A flexible molecule has quantum yield $\Phi = 0.25$ and lifetime $\tau = 5\ ns$.
If rigidification reduces internal conversion by half (k_{nr} halved), what will be the molecule's new Φ?
Q.13. If the radiative rate constant for a fluorophore is (k_r) = $2 \times 10^8\ s^{-1}$ and non-radiative rate constant is (k_{nr}) = $3 \times 10^8\ s^{-1}$, what is the fluorescence lifetime (τ) of the molecule?
Q.14. Upon excitation at 240 nm a molecule exhibits fluorescence photon emission at 210 nm. Is this statement correct? If not give reasons.
Q.15. Distinguish between flame photometry and UV–Visible absorption spectroscopy by highlighting at least four differences.